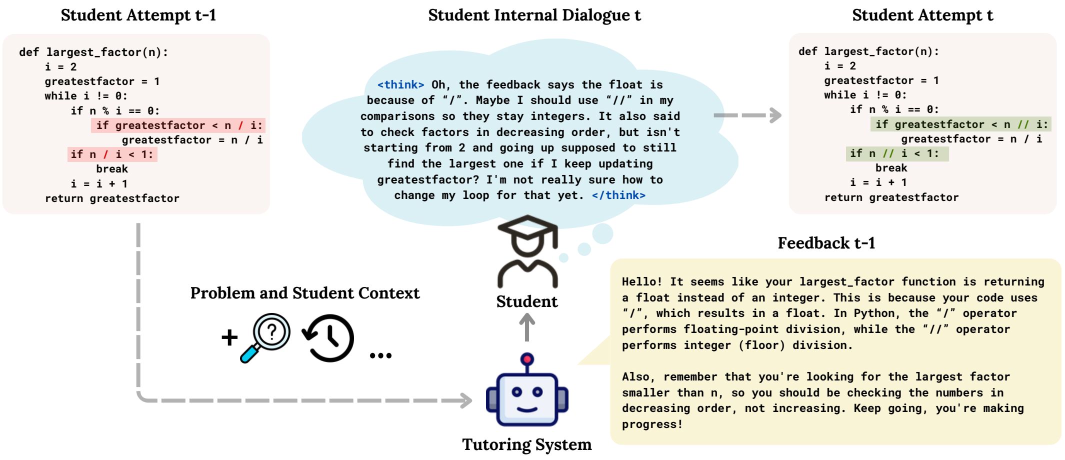
Student Attempt t-1
def largest_factor(n):
i = 2
greatestfactor = 1
while i != 0:
if n % i == 0:
if greatestfactor < n / i:
greatestfactor = n / i
if n / i < 1:
break
i = i + 1
return greatestfactor
Student Internal Dialogue t
<think> Oh, the feedback says the float is because of “/”. Maybe I should use “//” in my comparisons so they stay integers. It also said to check factors in decreasing order, but isn't starting from 2 and going up supposed to still find the largest one if I keep updating greatestfactor? I'm not really sure how to change my loop for that yet. </think>
Student Attempt t
def largest_factor(n):
i = 2
greatestfactor = 1
while i != 0:
if n % i == 0:
if greatestfactor < n // i:
greatestfactor = n / i
if n // i < 1:
break
i = i + 1
return greatestfactor
Student
Feedback t-1
Hello! It seems like your largest_factor function is returning a float instead of an integer. This is because your code uses “/”, which results in a float. In Python, the “/” operator performs floating-point division, while the “//” operator performs integer (floor) division.
Also, remember that you're looking for the largest factor smaller than n, so you should be checking the numbers in decreasing order, not increasing. Keep going, you're making progress!
Problem and Student Context
+
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Tutoring System